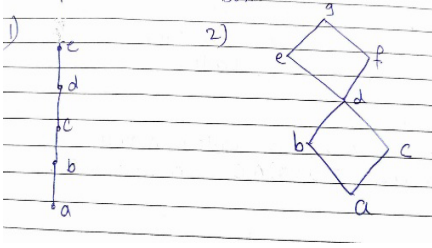
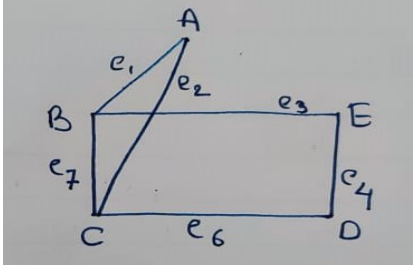
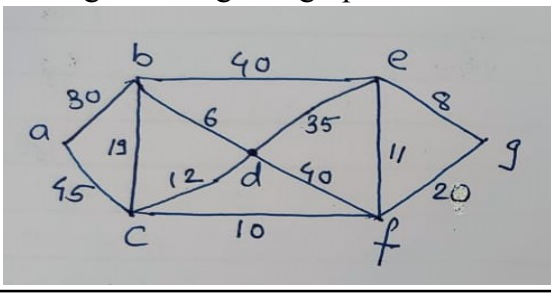
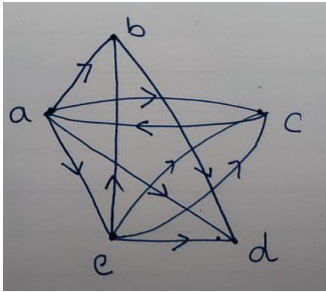
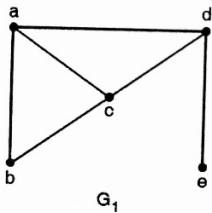
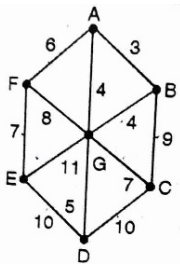
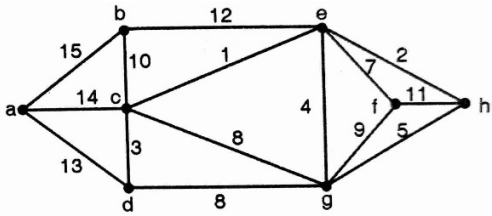
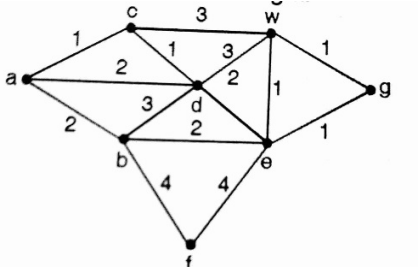
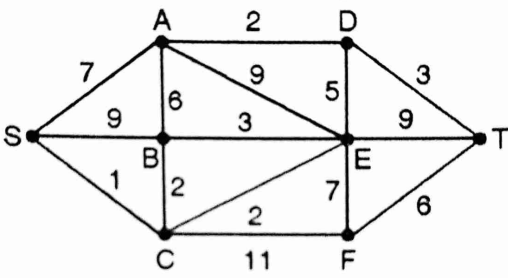


GHRCEM WAGHOLI PUNE
DEPARTMENT OF AI & AIML
SY CSE AI, AIML, CS & DS
QUESTION BANK FOR CAE 2
SUBJECT: DMGT (UBSL205)

Sr. No.	Question	Unit	BL	CO	PO
1	Let $A = \{1, 2, 3, 4, 6, 9, 12\}$ under divisibility relation prove that it is a partial order relation. Draw hasse diagram of the same. Prove or disprove if it is a lattice?	3	3	2	1
2	Let $S = \{2, 3, 6, 12, 24, 36\}$ with divisibility relation. $A = \{2, 3, 12\}$ $B = \{6, 12, 24, 36\}$ $C = \{3, 12\}$ are subsets of S . Find 1) minimal element 2) maximal element 3) least element 4) greatest element 5) ub, lub, lb, glb of the subsets A, B and C 6) Is S lattice?	3	5	2	1
3	Is every field is an Integral domain? If yes justify your answer Is every Integral domain is a feild? If yes justify your answer. If no give counter example	3	4	2	1
4	Is Z_8 integral domain? Justify.	3	4	2	1
5	a) Draw the Hasse diagram representing the partial ordering $\{(a, b): a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$ b) Draw the Hasse diagram for the less than or equal to relation on $\{0, 2, 5, 10, 11, 15\}$ c) Draw the Hasse diagram for the Poset $(p(S), \subseteq)$, where $p(S)$ is the power set of set $S = \{a, b, c\}$ d) Draw hasse diagram of $(D_{30}, ' ')$	3	3	2	2
6	a) Define commutative ring with unity b) Define Subring c) Define integral domain d) Define Field	3	2	2	2
7	Give an example of a ring without unity.	3	3	2	1

8	Show that $S=\{0,2,4,6\}$ is a subring of $(\mathbb{Z}_8, X_8, +_8)$	3	3	2	1
9	Determine whether each lattice is distributed, complemented or both. 	3	4	2	2
10	Let $L=D_{30}$ and $S_1= \{1, 3, 5, 15\}$, $S_2 = \{1, 6, 15, 30\}$. Check whether S_1 and S_2 are sub lattices of L or not.	3	4	2	2
11	Find adjacency matrix and incidence matrix of the following graph 	4	5	3	1
12	Using Dijkstra's algorithm find the shortest path between a and g in the given weighted graph 	4	5	3	1
13	Find adjacency matrix and incidence matrix of following diagram	4	5	3	1

					
14	Find adjacency matrix and incidence matrix of the following graph 	4	5	3	1
15	Draw the following graphs 1) K_7 2) $K_{4,5}$ 3) Bipartite graph on 8 vertices 4) 4-regular graph 5) Non-planar graph 6) Planar graph 7) 4-chromatic graph	4	2	3	1
16	Find number of vertices of a graph in which 3 vertices of degree 4, and all other vertices of degree 2 with exactly 21 edges.	4	3	3	2
17	Construct minimum spanning tree using Prim's algorithm. Obtain total cost of minimum spanning tree 	4	5	3	1
18	Find Minimum spanning tree using kruskal's algorithm. Obtain total cost of minimum spanning tree. 	4	5	3	1

19	Construct minimum spanning tree using Prim's algorithm, starting at vertex a. Obtain total cost of minimum spanning tree 	4	5	3	1
20	Find Minimum spanning tree using kruskal's algorithm. Obtain total cost of minimum spanning tree. 	4	5	3	1

Shaheen Shaikh/Suvarna Gaikwad

SUBJECT INCHARGE

HOD